Problem-Solving, Ratios, Rates & Data

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

According to the bar chart shown for the community theater, what was the total number of visitors across all four quarters?

Answer: 90. Reading the bar heights from the bar chart: Quarter 1 has 16, Quarter 2 has 24, Quarter 3 has 19, and Quarter 4 has 31. The total is $16 + 24 + 19 + 31 = 90$.

2 [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

According to the bar chart shown for the mountain observatory, how many more visitors were recorded in Quarter 4 than in Quarter 1?

Answer: 18. Reading the bar heights from the bar chart: Quarter 4 has 40 and Quarter 1 has 22. The difference is $40 - 22 = 18$.

3 [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

According to the bar chart shown for the solar microgrid facility, what was the total number of visitors across all four quarters?

Answer: 150. Reading the bar heights from the bar chart: Quarter 1 has 28, Quarter 2 has 42, Quarter 3 has 31, and Quarter 4 has 49. The total is $28 + 42 + 31 + 49 = 150$.

4 [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

According to the bar chart shown for the publishing house, how many more visitors were recorded in Quarter 4 than in Quarter 1?

Answer: 24. Reading the bar heights from the bar chart: Quarter 4 has 58 and Quarter 1 has 34. The difference is $58 - 34 = 24$.

5 [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

According to the bar chart shown for the hydroponic greenhouse, what was the total number of visitors across all four quarters?

Answer: 210. Reading the bar heights from the bar chart: Quarter 1 has 40, Quarter 2 has 60, Quarter 3 has 43, and Quarter 4 has 67. The total is $40 + 60 + 43 + 67 = 210$.

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6 [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

According to the bar chart shown for the robotics club, how many more visitors were recorded in Quarter 4 than in Quarter 1?

Answer: 30. Reading the bar heights from the bar chart: Quarter 4 has 76 and Quarter 1 has 46. The difference is $76 - 46 = 30$.

7 [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

According to the bar chart shown for the wildlife sanctuary, what was the total number of visitors across all four quarters?

Answer: 270. Reading the bar heights from the bar chart: Quarter 1 has 52, Quarter 2 has 78, Quarter 3 has 55, and Quarter 4 has 85. The total is $52 + 78 + 55 + 85 = 270$.

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the orchestra hall, if a car uses 4 gallons of gasoline to travel 120 miles, how many gallons would it use to travel 300 miles at the same rate?

Answer: 10. Fuel economy is $120/4 = 30$ mpg. For 300 miles: $300/30 = 10$ gallons.

9 [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the honeybee apiary, a printer prints 45 pages in 3 minutes. At this rate, how many pages can it print in 8 minutes?

Answer: 120. Rate = $45/3 = 15$ pages per minute. In 8 minutes: $15 \times 8 = 120$ pages.

10 [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the performing arts academy, if 6 workers take 8 days to complete a job, how many days would 12 workers take working at the same pace?

Answer: 4. Total worker-days = $6 \times 8 = 48$. With 12 workers: $48/12 = 4$ days.

11 [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the coastal aquarium, a line of best fit relating hours studied (x) to test score ($\hat{y}$) is $\hat{y} = 10x + 27$. Using this model, what is the predicted score for a student who studies 4 hours?

Answer: 67. Substitute $x = 4$ into the regression equation: $\hat{y} = 10(4) + 27 = 40 + 27 = 67$.

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12 [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the wildlife sanctuary, a company increased its workforce by 25% this year. If the new total is 250 employees, how many employees did the company have before the increase?

Answer: 200. $1.25E = 250 \Rightarrow E = 250/1.25 = 200$.

13 [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the agricultural cooperative, a class of 26 students has an average score of 63, and another class of 26 students has an average score of 71. What is the average score of all students combined?

Answer: 67. Combined average $= \frac{26(63)+26(71)}{26+26} = \frac{3484}{52} = 67$

14 [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the wildlife sanctuary, an item's price increased from USD 50 to USD 65. What was the percent increase?

Answer: 2. Percent increase = $\frac{65-50}{50} \times 100\% = \frac{15}{50} \times 100\% = 30\%$.

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the historic lighthouse, a car's fuel tank is $\frac{1}{4}$ full. After adding 15 gallons of fuel, the tank is $\frac{7}{8}$ full. What is the total capacity of the fuel tank in gallons?

Answer: 24. Fraction added = $\frac{7}{8} - \frac{1}{4} = \frac{5}{8}$. Let C be capacity: $\frac{5}{8}C = 15 \Rightarrow C = 15 \times \frac{8}{5} = 24$ gallons.

16 [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

Last year, the botanical conservatory sold 200 adult tickets. This year, sales of adult tickets increased by 10% over last year. How many adult tickets did the botanical conservatory sell this year?

Answer: 220. An increase of 10% multiplies the original quantity by $1 + 0.1 = 1.10$. Thus $200 \times 1.10 = 220$.

17 [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A packaging machine at the municipal library seals hardcover books at a constant rate of 16 units per minute. At this rate, how many units does the machine seal in 3 hours?

Answer: 2880. Convert 3 hours to minutes: $3 \times 60 = 180$ minutes. Multiplying by the rate of 16 units per minute gives $16 \times 180 = 2880$ units.

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18 [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

At the community theater, the ratio of small planters to large planters is 5 to 7. If there are a total of 96 of these items combined, how many are small planters?

Answer: 40. The ratio has $5 + 7 = 12$ total parts. Each part represents $\frac{96}{12} = 8$ items, so there are $5 \times 8 = 40$ small planters.

19 [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

Last year, the regional bakery sold 500 standard passes. This year, sales of standard passes increased by 40% over last year. How many standard passes did the regional bakery sell this year?

Answer: 700. An increase of 40% multiplies the original quantity by $1 + 0.4 = 1.40$. Thus $500 \times 1.40 = 700$.

20 [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A packaging machine at the coastal aquarium seals oak saplings at a constant rate of 19 units per minute. At this rate, how many units does the machine seal in 3 hours?

Answer: 3420. Convert 3 hours to minutes: $3 \times 60 = 180$ minutes. Multiplying by the rate of 19 units per minute gives $19 \times 180 = 3420$ units.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

21 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

According to the bar chart shown for the botanical conservatory, what was the total number of visitors across all four quarters?

a. 70 ✓

b. 76

c. 66

d. 33

Reading the bar heights from the bar chart: Quarter 1 has 12, Quarter 2 has 18, Quarter 3 has 15, and Quarter 4 has 25. The total is $12 + 18 + 15 + 25 = 70$.

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22 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

According to the bar chart shown for the municipal library, how many more visitors were recorded in Quarter 4 than in Quarter 1?

a. 20

b. 10

c. 38

d. 14 ✓

Reading the bar heights from the bar chart: Quarter 4 has 28 and Quarter 1 has 14. The difference is $28 - 14 = 14$.

23 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

According to the bar chart shown for the regional bakery, how many more visitors were recorded in Quarter 4 than in Quarter 1?

a. 48

b. 16 ✓

c. 22

d. 12

Reading the bar heights from the bar chart: Quarter 4 has 34 and Quarter 1 has 18. The difference is $34 - 18 = 16$.

24 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

According to the bar chart shown for the coastal aquarium, what was the total number of visitors across all four quarters?

a. 110 ✓

b. 116

c. 106

d. 53

Reading the bar heights from the bar chart: Quarter 1 has 20, Quarter 2 has 30, Quarter 3 has 23, and Quarter 4 has 37. The total is $20 + 30 + 23 + 37 = 110$.

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25 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

According to the bar chart shown for the science museum, what was the total number of visitors across all four quarters?

a. 126

b. 63

c. 130 ✓

d. 136

Reading the bar heights from the bar chart: Quarter 1 has 24, Quarter 2 has 36, Quarter 3 has 27, and Quarter 4 has 43. The total is $24 + 36 + 27 + 43 = 130$.

26 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

According to the bar chart shown for the urban bike-share station, how many more visitors were recorded in Quarter 4 than in Quarter 1?

a. 68

b. 20 ✓

c. 26

d. 16

Reading the bar heights from the bar chart: Quarter 4 has 46 and Quarter 1 has 26. The difference is $46 - 26 = 20$.

27 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

According to the bar chart shown for the ceramic studio, how many more visitors were recorded in Quarter 4 than in Quarter 1?

a. 28

b. 18

c. 78

d. 22 ✓

Reading the bar heights from the bar chart: Quarter 4 has 52 and Quarter 1 has 30. The difference is $52 - 30 = 22$.

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28 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

According to the bar chart shown for the agricultural cooperative, what was the total number of visitors across all four quarters?

a. 166

b. 83

c. 170 ✓

d. 176

Reading the bar heights from the bar chart: Quarter 1 has 32, Quarter 2 has 48, Quarter 3 has 35, and Quarter 4 has 55. The total is $32 + 48 + 35 + 55 = 170$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the charter ferry service, a runner completes a 5-kilometer race in 25 minutes. What is the runner's average speed in kilometers per hour?

a. 10 km/h

b. **12 km/h ✓**

c. 15 km/h

d. 20 km/h

25 minutes is $25/60 = 5/12$ hours. Speed $= 5/(5/12) = 12$ km/h.

30 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the textile workshop, if x is directly proportional to y, and $x = 14$ when $y = 2$, what is x when $y = 5$?

a. 25

b. 30

c. 35 ✓

d. 40

Constant of proportionality $k = x/y = 14/2 = 7$. When $y = 5$, $x = 7 \times 5 = 35$.

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31 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the geology lab, an electronic device is discounted from USD 200 to USD 160. What is the percent discount?

a. 15%

b. 20% ✓

c. 25%

d. 40%

$\frac{200-160}{200} \times 100\% = \frac{40}{200} \times 100\% = 20\%$.

32 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the carpentry workshop, if a 12-ounce beverage contains 150 calories, how many calories are in an 8-ounce serving of the same beverage?

a. 90

b. 100 ✓

c. 110

d. 120

Calories per ounce = $150/12 = 12.5$. For 8 ounces: $12.5 \times 8 = 100$ calories.

33 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the historic lighthouse, a car travels 180 miles in 3 hours. At this same rate, how far will it travel in 5 hours?

a. 240 miles

b. 270 miles

c. **300 miles** ✓

d. 360 miles

Speed $= 180/3 = 60$ mph. In 5 hours: $60 \times 5 = 300$ miles.

34 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a calculation for the community orchard (case 19), if $\frac{a}{15} = \frac{4}{5}$, what is the value of a?

a. 10

b. 12 ✓

c. 14

d. 16

$a = 15 \times \frac{4}{5} = 12$.

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35 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the local farmers market, there are 1,000 milliliters in 1 liter. How many milliliters are in 3.4 liters?

a. 340 mL

b. **3,400 mL** ✓

c. 34,000 mL

d. 34 mL

$3.4 \times 1000 = 3,400$ mL.

36 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the environmental club, a store sells notebooks in packs of 5 for USD 6. How much would 25 notebooks cost at this rate?

a. USD 24

b. **USD** 30 ✓

c. USD 35

d. USD 40

25 notebooks is 5 packs: $5 \times USD6 = USD30$.

37 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the neighborhood hardware store, if a team won 21 games and lost 7 games, what fraction of their games did they win?

a. $\frac{1}{3}$

b. $\frac{2}{3}$

c. $\frac{3}{4}$ ✓

d. $\frac{4}{5}$

Total games = $21 + 7 = 28$. Fraction won = $21/28 = 3/4$.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the honeybee apiary, if a price is increased by x% and then decreased by x%, the final price is USD 384. If the original price was USD 400, what is the value of x?

a. 10

b. 15

c. 20 ✓

d. 25

$400\left(1 - \frac{x^2}{10000}\right) = 384 \Rightarrow 1 - \frac{x^2}{10000} = \frac{384}{400} = 0.96 \Rightarrow \frac{x^2}{10000} = 0.04 \Rightarrow x^2 = 400 \Rightarrow x = 20.$

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39 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the carpentry workshop, two investment funds started with USD 10,000 each. Fund A increased by 10% in Year 1 and 20% in Year 2. Fund B increased by 15% in both Year 1 and Year 2. What is the positive difference in their balances after Year 2?

a. **USD 25** ✓

b. USD 50

c. USD 75

d. USD 100

Fund A: $10,000(1.10)(1.20) = 10,000(1.32) = USD13,200$. Fund B: $10,000(1.15)(1.15) = 10,000(1.3225) = USD13,225$. Difference = $13,225 - 13,200 = USD25$.

40 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the community orchard, a company's production cost is modeled by $C(x) = 5000 + 45x$, where x is units produced. If each unit sells for USD 70, how many units must be sold to make a profit of USD 10,000?

a. 400

b. 500

c. 600 ✓

d. 750

Profit = Revenue - Cost: $70x - (5000 + 45x) = 10,000 \Rightarrow 25x - 5000 = 10,000 \Rightarrow 25x = 15,000 \Rightarrow x = 600$.

41 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

In a model for the local farmers market, a high school has 400 seniors. 180 take AP Calculus, 140 take AP Physics, and 80 take both. How many seniors take neither AP Calculus nor AP Physics?

a. 120

b. 140

c. 160 ✓

d. 180

By principle of inclusion-exclusion: taking at least one = $180 + 140 - 80 = 240$. Taking neither = $400 - 240 = 160$.

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42 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

Last year, the charter ferry service sold 200 hiking packs. This year, sales of hiking packs increased by 10% over last year. How many hiking packs did the charter ferry service sell this year?

a. 220 ✓

b. 210

c. 200

d. 250

An increase of 10% multiplies the original quantity by $1 + 0.1 = 1.10$. Thus $200 \times 1.10 = 220$.

43 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A packaging machine at the maple syrup farm seals chess sets at a constant rate of 16 units per minute. At this rate, how many units does the machine seal in 3 hours?

a. 48

b. 1440

c. 2940

d. 2880 ✓

Convert 3 hours to minutes: $3 \times 60 = 180$ minutes. Multiplying by the rate of 16 units per minute gives $16 \times 180 = 2880$ units.

44 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

At the textile workshop, the ratio of microscope slides to glass beakers is 4 to 6. If there are a total of 80 of these items combined, how many are microscope slides?

a. 8

b. 36

c. 32 ✓

d. 48

The ratio has $4 + 6 = 10$ total parts. Each part represents $\frac{80}{10} = 8$ items, so there are $4 \times 8 = 32$ microscope slides.

45 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

Last year, the planetarium sold 500 paint brushes. This year, sales of paint brushes increased by 40% over last year. How many paint brushes did the planetarium sell this year?

a. 730

b. 700 ✓

c. 540

d. 680

An increase of 40% multiplies the original quantity by $1 + 0.4 = 1.40$. Thus $500 \times 1.40 = 700$.

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